

Interference of resonances in two-proton emission of ^{16}Ne

Journal club, Chenrong Ding

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- Two-proton(2p) radioactivity is a characteristic decay process beyond the proton-dripline. This phenomenon serves as a natural laboratory to investigate the open-quantum systems, quantum entanglement, and pairing correlations in nuclear systems.
- There are two mechanisms of 2p emission: simultaneous and sequential emissions.
- The ^{16}Ne nucleus is a 2p emitter, which can be approximated as the two valence protons around the proton shell-closed core ^{14}O ($Z=8$). the $s_{1/2}$ resonance in ^{15}F can play a role of so-called intermediate resonance to invoke the sequential emission.
- This paper aims to clarify (i) whether the 0_2^+ resonance exists or not in ^{16}Ne , and (ii) how it affects the 2p emitting process.

➤ The three-body (3B) Hamiltonian

$$\hat{H}_{3B} = \hat{h}(\mathbf{r}_1) + \hat{h}(\mathbf{r}_2) + v_{pp}(\mathbf{r}_1, \mathbf{r}_2) + \frac{\mathbf{p}_1 \cdot \mathbf{p}_2}{m_C} + \frac{\mathbf{P}_{CM}^2}{2M}.$$

- $\hat{h}(r_i)$ is the Hamiltonian for the subsystem composed of the i-th valence proton and the core:

$$\hat{h}(r_i) = -\frac{\hbar^2}{2\mu} \frac{d^2}{dr_i^2} + V(r_i),$$

$$V(r_i) = \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r_i^2} + \underbrace{V_{WS}(r_i)}_{\text{Spherical Woods-Saxon potential}} + \underbrace{V_{Coul}(r_i)}_{\text{Coulomb potential}},$$

Spherical Woods-Saxon potential

Coulomb potential

$$V_{WS}(r) = V_0 f(r) + U_{ls}(\mathbf{l} \cdot \mathbf{s}) \frac{1}{r} \frac{df(r)}{dr},$$

$$f(r) = \frac{1}{1 + e^{(r-R_0)/a_0}},$$

$$V_{Coul}(r) = \begin{cases} \hbar c \frac{Z\alpha_0}{r} & (r \geq R_0) \\ \hbar c \frac{Z\alpha_0}{2R_0} \left[3 - \left(\frac{r}{R_0} \right)^2 \right] & (r < R_0) \end{cases}$$

- $v_{pp}(r_1, r_2)$ is the proton-proton interactions:

$$v_{pp}(\mathbf{r}_1, \mathbf{r}_2) = v_{pp,vac}(\mathbf{r}_1, \mathbf{r}_2) + v_{pp,add}(\mathbf{r}_1, \mathbf{r}_2).$$

$$v_{pp,vac}(\mathbf{r}_1, \mathbf{r}_2) = \left[V_R e^{-a_R r^2} + V_S e^{-a_S r^2} \hat{P}_{S=0} + V_T e^{-a_T r^2} \hat{P}_{S=1} \right] + v_{pp,Coul}(r),$$

$$v_{pp,add}(\mathbf{r}_1, \mathbf{r}_2) = w_0 e^{-(R-R_0)^2/B_0^2} \delta(\mathbf{r}_1 - \mathbf{r}_2),$$

➤ The three-body (3B) Hamiltonian

$$\hat{H}_{3B} = \hat{h}(\mathbf{r}_1) + \hat{h}(\mathbf{r}_2) + v_{pp}(\mathbf{r}_1, \mathbf{r}_2) + \frac{\mathbf{p}_1 \cdot \mathbf{p}_2}{m_C} + \frac{\mathbf{P}_{CM}^2}{2M}.$$

TABLE II. Core-proton $s_{1/2}$ and $d_{5/2}$ resonances of the $^{15}\text{F} = ^{14}\text{O}+p$. The unit is MeV.

^{15}F	$s_{1/2}$		$d_{5/2}$	
	$E_p(s_{1/2})$	$\Gamma_p(s_{1/2})$	$E_p(d_{5/2})$	$\Gamma_p(d_{5/2})$
Expt.[21]	1.23(5)	0.50-0.84	2.81(2)	0.30(6)
[20]	1.23-1.37	0.7	2.795(45)	0.325(6)
	1.35-1.61	—		
[19]	1.51(15)	1.2	2.853(45)	0.34
[18]	1.47(13)	1.0(2)	2.77(10)	0.24(3)
prompt	$\cong 1.5$	$\cong 1.3$	2.785	0.391
mixed	1.277	0.604	2.787	0.264

Mixed

Prompt

Fitting

$$V_{WS}(r) = V_0 f(r) + U_{ls}(\mathbf{l} \cdot \mathbf{s}) \frac{1}{r} \frac{df(r)}{dr},$$

$$f(r) = \frac{1}{1 + e^{(r-R_0)/a_0}},$$

TABLE I. Summary of model parameters.

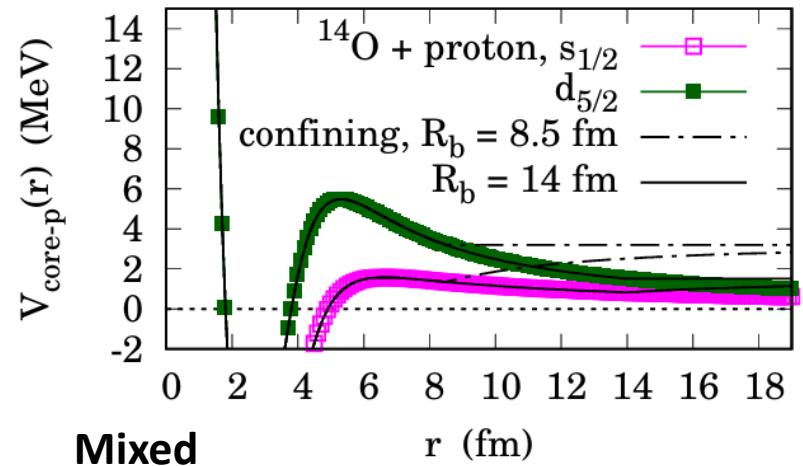
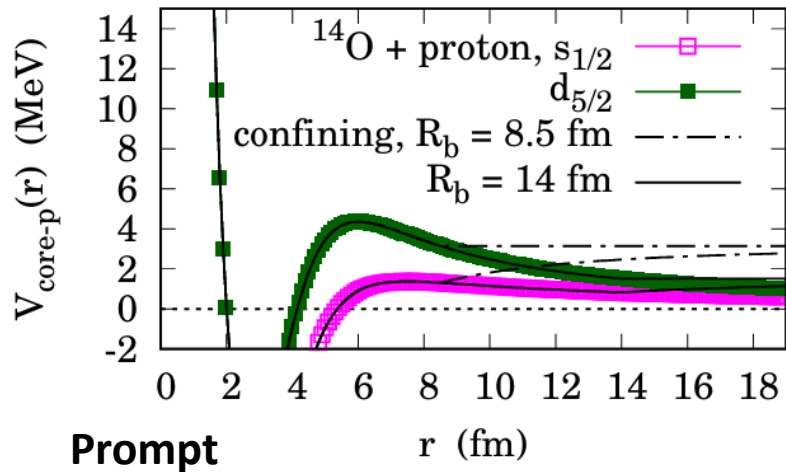
		prompt	mixed
core-proton	r_0 [fm]	1.23	1.23
	a_0 [fm]	0.64	0.82
	V_0 [MeV]	-42.25	-51.53
	U_{ls} [MeV·fm ²]	54.1	18.68
proton-proton	w_0 [MeV·fm ³]	-527	-260

➤ The three-body (3B) Hamiltonian

$$\hat{H}_{3B} = \hat{h}(\mathbf{r}_1) + \hat{h}(\mathbf{r}_2) + v_{pp}(\mathbf{r}_1, \mathbf{r}_2) + \frac{\mathbf{p}_1 \cdot \mathbf{p}_2}{m_C} + \frac{\mathbf{P}_{CM}^2}{2M}.$$

$$\langle \Psi_{n_1 l_1 j_1 n_2 l_2 j_2}^J \| H \| \Psi_{n'_1 l'_1 j'_1 n'_2 l'_2 j'_2}^J \rangle = \langle \Psi_{n_1 n_2 l j}^0 \| H \| \Psi_{n'_1 n'_2 l' j'}^0 \rangle, \quad (J = 0, l_1 j_1 = l_2 j_2 = l j, l'_1 j'_1 = l'_2 j'_2 = l' j').$$

For simulating the emission, two protons should be phenomenologically confined inside the potential barrier of the core. To prepare such an initial 2p state:



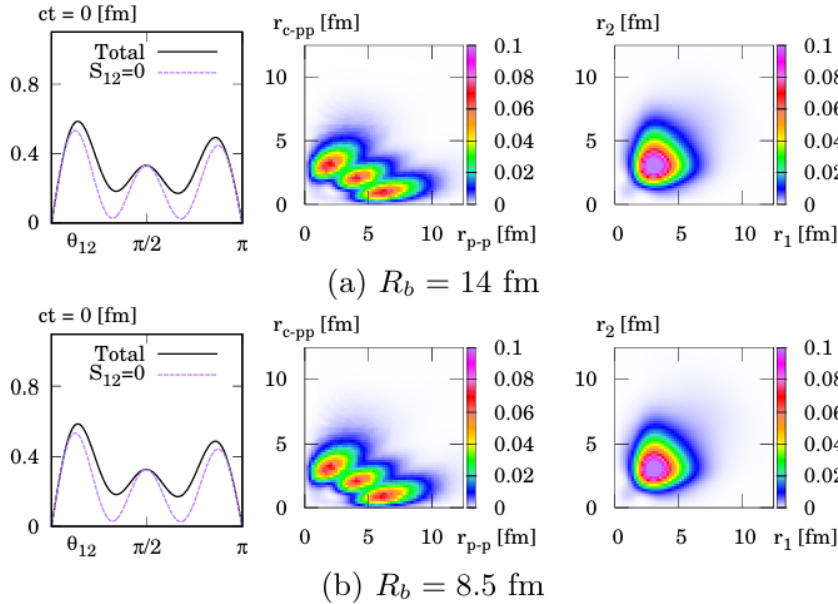


FIG. 4. (a) Density distribution of 2p state at $t = 0$ for ^{16}Ne in the prompt case with $R_b = 14$ fm. For convention of plotting variables, see Refs. [5, 37, 38]. (b) Same but by changing the confining radius as $R_b = 8.5$ fm.

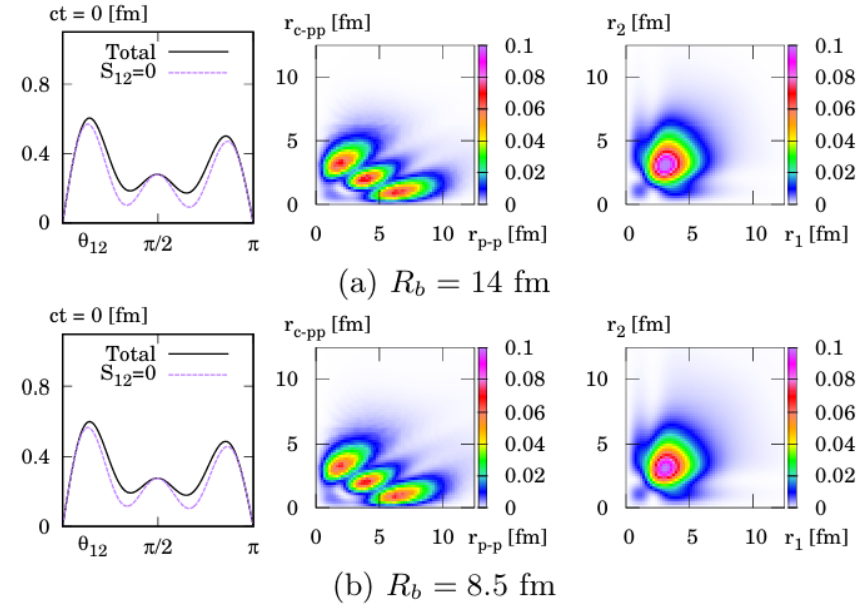
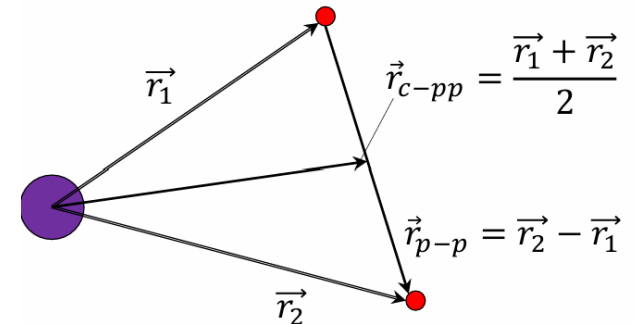


FIG. 5. Same to FIG. 4 but in the mixed case.

- The 2p are confined around the core.
- For both of the prompt and mixed cases, the initial 2p density distributions show a similar profile.
- The angular asymmetric form is a product of the mixture of ($l \in \text{even}$) and ($l \in \text{odd}$) configurations.



When the initial state is fixed, one can expand it as $|\Psi(0)\rangle = \sum_N F_N(0) |E_N\rangle$, and its **time development** is solved as

$$|\Psi(t)\rangle = \exp\left[-it\frac{\hat{H}_{3B}}{\hbar}\right] |\Psi(0)\rangle = \sum_N F_N(t) |E_N\rangle, \quad (15)$$

where $F_N(t) = e^{-it\frac{E_N}{\hbar}} F_N(0)$. **Time-evolution probability** is determined as $P_T(t) = |\beta(t)|^2$, where [59, 60]

$$\beta(t) = \langle \Psi(0) | \Psi(t) \rangle = \sum_N e^{-it\frac{E_N}{\hbar}} |F_N(0)|^2. \quad (16)$$

Note that $\beta(t) = \int e^{-it\frac{E}{\hbar}} |F(E, t=0)|^2 dE$ in the continuous limit. The **mean-energy release**, $Q_{2p} = -S_{2p}$, is evaluated as

$$Q_{2p} = \langle \Psi(t) | \hat{H}_{3B} | \Psi(t) \rangle = \sum_N |F_N(0)|^2 E_N. \quad (17)$$

Notice that this quantity is time-independent.

If the decay process was governed by a single-peak spectrum:

$$P_T(t) \cong e^{-t/\tau} \quad \Gamma = \hbar/\tau$$

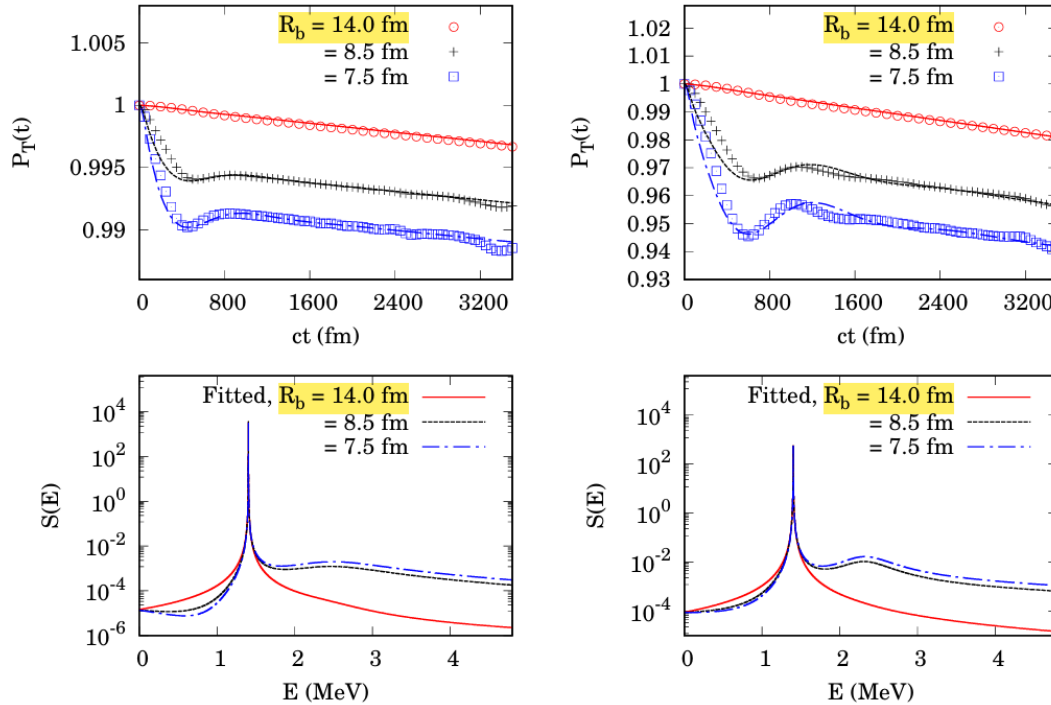
Thus, the corresponding energy distribution has the single Breit-Wigner profile:

$$F(E) = L_1(E) = \sqrt{\frac{\Gamma_1}{2\pi}} \frac{1}{E - (E_1 - i\Gamma_1/2)}.$$

And the energy spectra:

$$S(E) = |F(E)|^2$$

Methods: time-dependent calculations



	k	1 for 0_1^+
prompt	E_k [MeV]	1.404
	Γ_k [MeV]	1.4×10^{-4}
	$u_2(R_b = 8.5 \text{ fm})$	
	$u_2(R_b = 7.5 \text{ fm})$	
	$(s_{1/2})^2$ [%]	8.5
	$(d_{5/2})^2$ [%]	88.4
mixed	E_k [MeV]	1.401
	Γ_k [MeV]	1.1×10^{-3}
	$u_2(R_b = 8.5 \text{ fm})$	
	$u_2(R_b = 7.5 \text{ fm})$	
	$(s_{1/2})^2$ [%]	32.0
	$(d_{5/2})^2$ [%]	63.5

If the decay process was governed by a single-peak spectrum:

$$P_T(t) \cong e^{-t/\tau} \quad \Gamma = \hbar/\tau$$

Thus, the corresponding energy distribution has the single Breit-Wigner profile:

$$F(E) = L_1(E) = \sqrt{\frac{\Gamma_1}{2\pi}} \frac{1}{E - (E_1 - i\Gamma_1/2)}.$$

And the energy spectra:

$$S(E) = |F(E)|^2$$

Results: first 0+ resonance

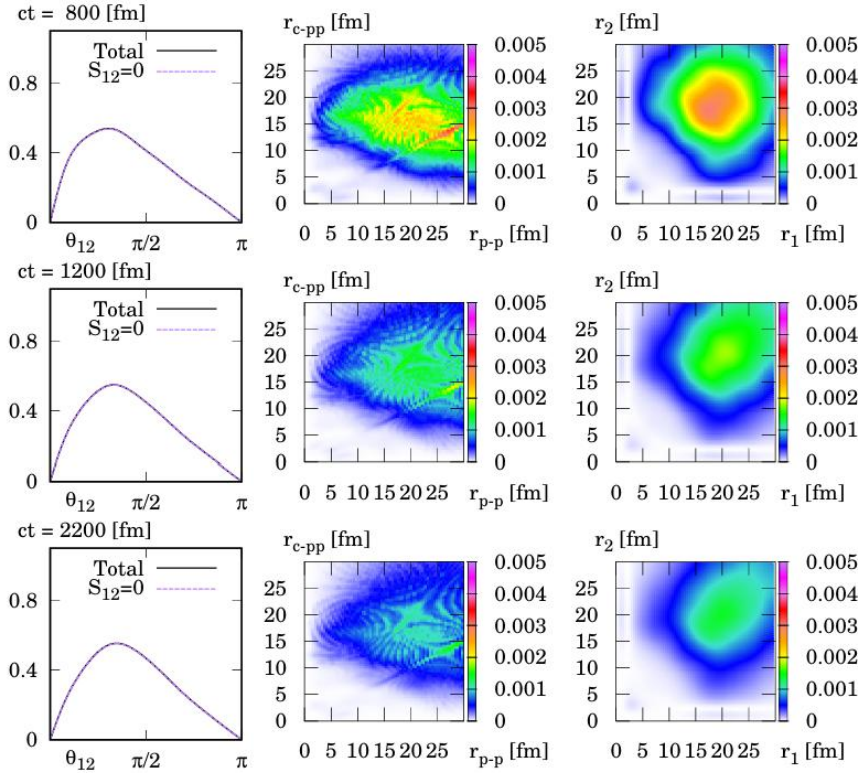


FIG. 8. Density distributions of the time-dependent decaying $2p$ state: see Eqs. (19) and (20) for definition. Results in the prompt case with $R_b = 14$ fm.

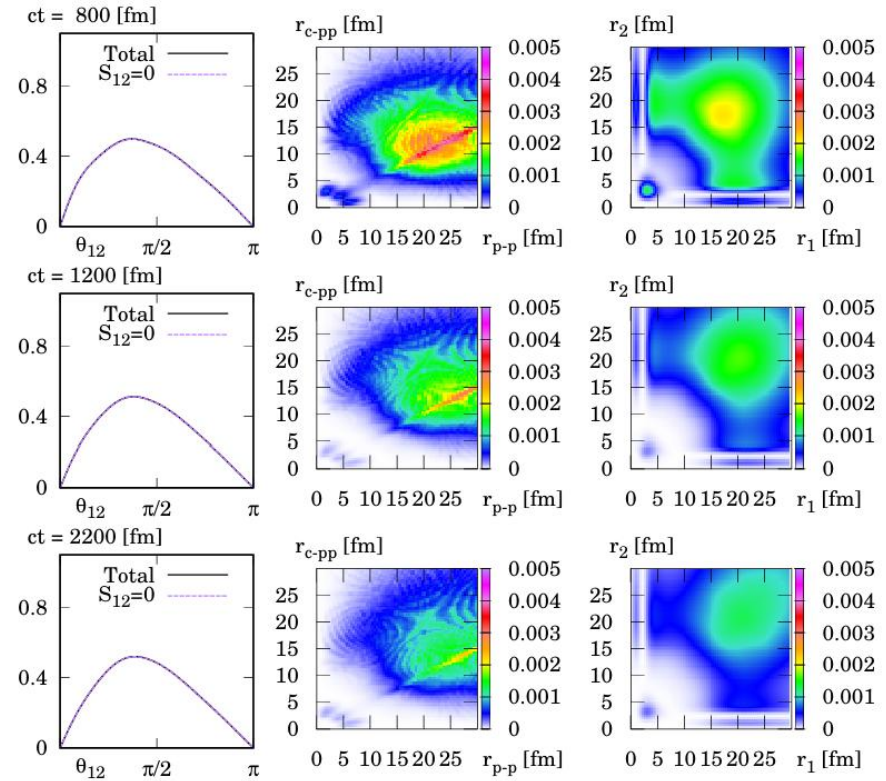
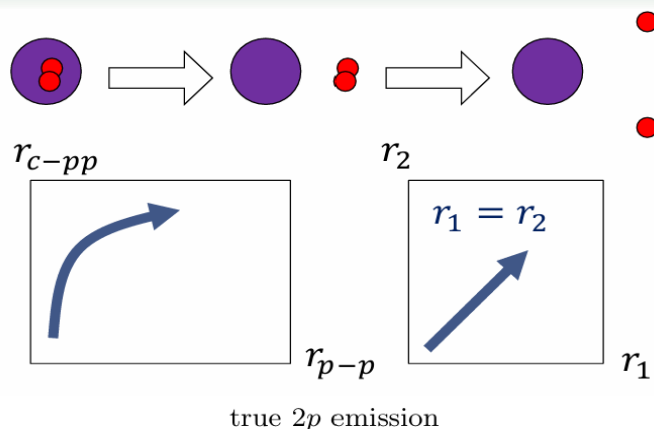
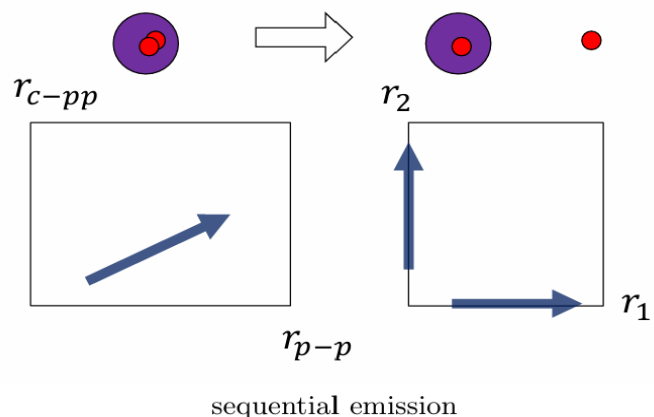


FIG. 10. Same to FIG. 8 but in the mixed case with $R_b = 14$ fm.

Results: first 0+ resonance



The prompt case exhibits the true 2p emission as dominant process, which is characterized by the asymmetric opening-angle distribution and the profile in the $r_1=r_2$ region.



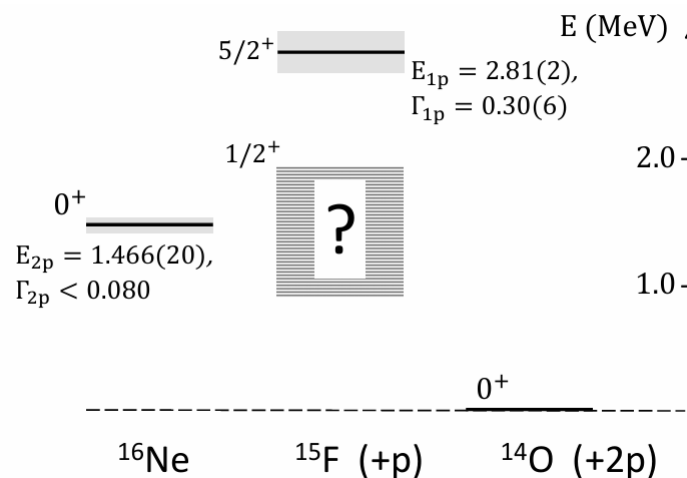
In the mixed case, in contrast, a sequential 1p-1p emission becomes apparent. This appears as the trajectory along the $r_c-pp=r_p-p/2$ line.

¹⁵ F	$s_{1/2}$		$d_{5/2}$	
	$E_p(s_{1/2})$	$\Gamma_p(s_{1/2})$	$E_p(d_{5/2})$	$\Gamma_p(d_{5/2})$
Expt.[21]	1.23(5)	0.50-0.84	2.81(2)	0.30(6)
[20]	1.23-1.37	0.7	2.795(45)	0.325(6)
	1.35-1.61	—		
[19]	1.51(15)	1.2	2.853(45)	0.34

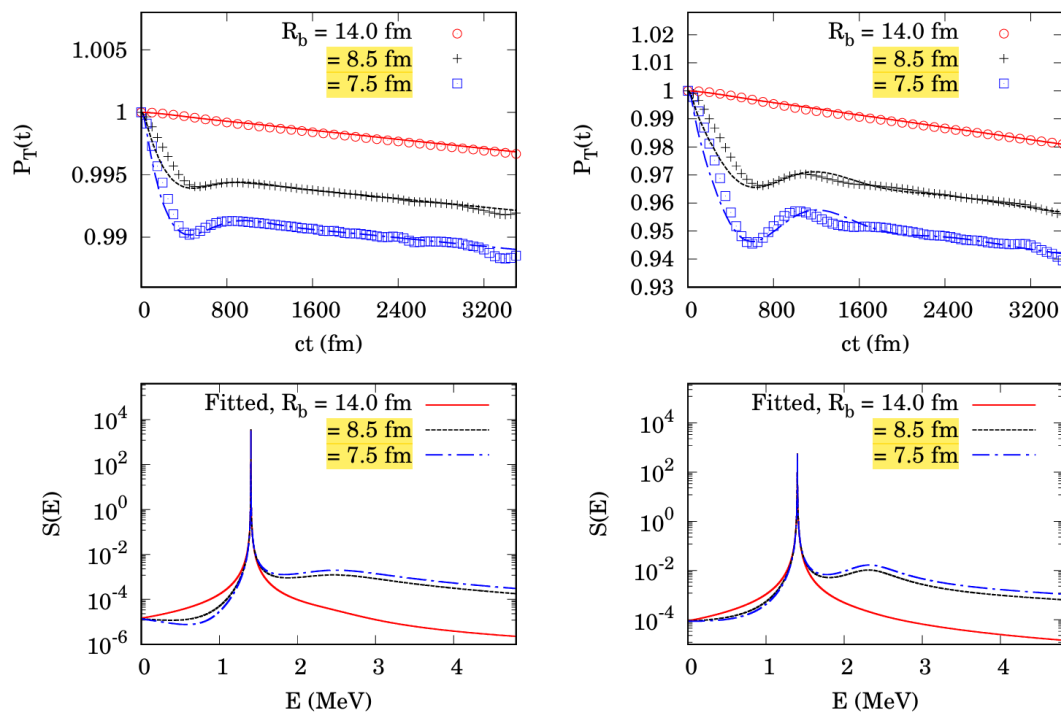
$$E_k \approx 1.4 \text{ MeV}$$

Mixed

Prompt



Methods: time-dependent calculations



The rapid decrease of $P_T(t)$ followed by oscillation until $ct \sim 1200$ fm represents the mixture of 0_2^+ resonance.

Thus, the corresponding energy distribution:

$$F(E) = \sqrt{N_{12}} [L_1(E) + u_2 L_2(E)]^2,$$

where

$$L_2(E) = \frac{1}{\sqrt{N_2} E - (E_2 - i\Gamma_2(E)/2)},$$

$$\Gamma_2(E) = \Gamma_2 \left[0.65 \left(\frac{E}{E_2} \right)^2 + 0.35 \left(\frac{E}{E_2} \right)^4 \right].$$

And the energy spectra:

$$S(E) = |F(E)|^2$$

	k	1 for 0_1^+	2 for 0_2^+
prompt	E_k [MeV]	1.404	2.96
	Γ_k [MeV]	1.4×10^{-4}	2.77
	$u_2(R_b = 8.5 \text{ fm})$		-0.047
	$u_2(R_b = 7.5 \text{ fm})$		-0.061
	$(s_{1/2})^2$ [%]	8.5	70.1
	$(d_{5/2})^2$ [%]	88.4	5.7
mixed	E_k [MeV]	1.401	2.44
	Γ_k [MeV]	1.1×10^{-3}	0.82
	$u_2(R_b = 8.5 \text{ fm})$		-0.109
	$u_2(R_b = 7.5 \text{ fm})$		-0.140
	$(s_{1/2})^2$ [%]	32.0	61.2
	$(d_{5/2})^2$ [%]	63.5	33.7

Results: second 0+ resonance

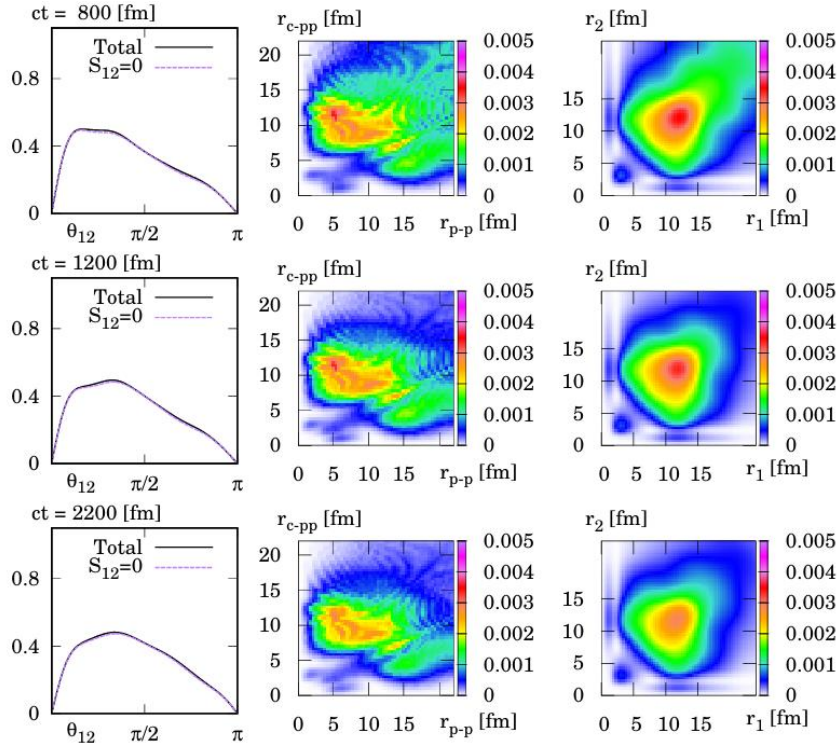


FIG. 9. Same to FIG. 8 (prompt) but $R_b = 8.5$ fm.

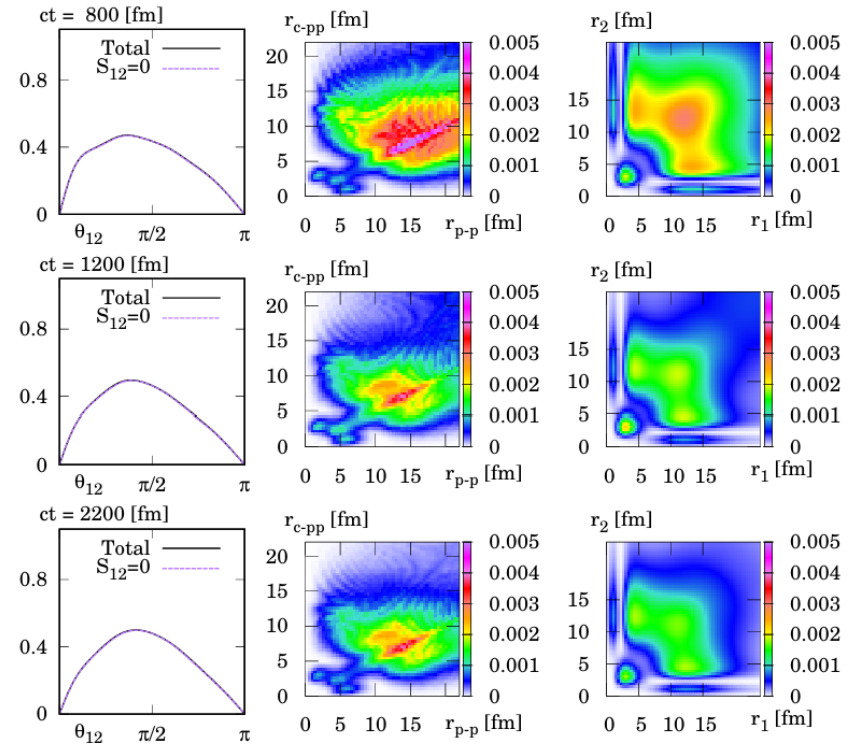


FIG. 11. Same to FIG. 10 (mixed) but $R_b = 8.5$ fm.

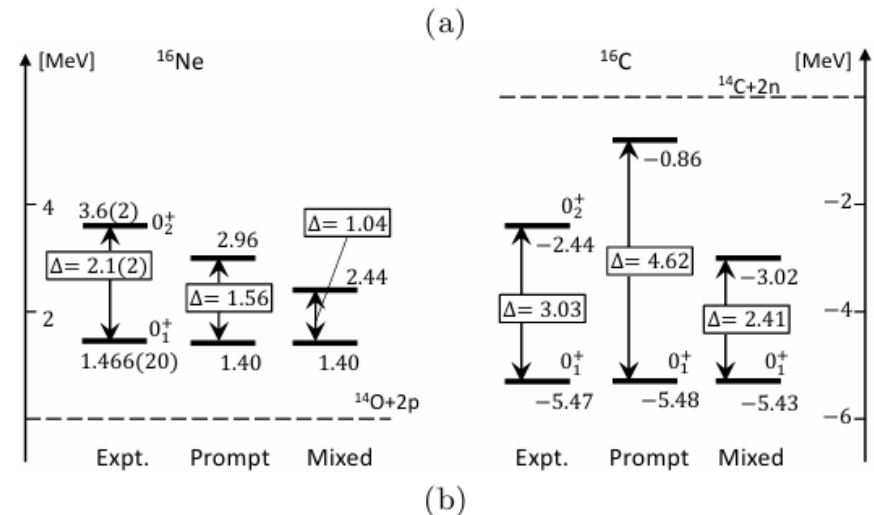
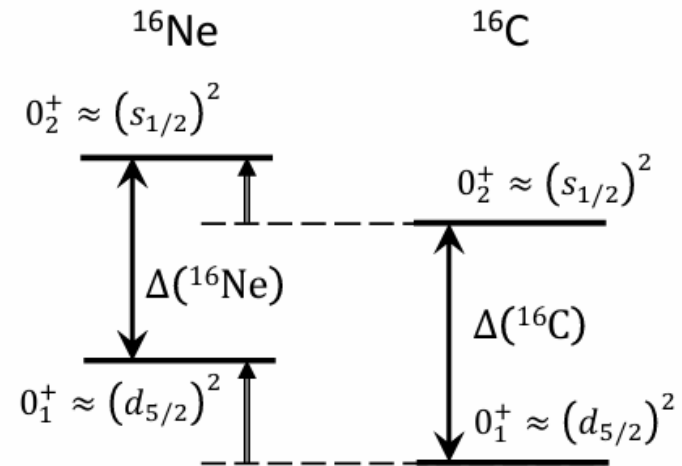
By comparing the FIGs.8 and 9 in the prompt case, an inclusion of the 0_2^+ resonance helps the prompt-2p emission to become more apparent.

Results: Coulomb shift in ^{16}Ne and ^{16}C



The coulomb force generally increase more the $d_{5/2}$ level, whereas the $s_{1/2}$ level is less sensitive.

	k	1 for 0_1^+	2 for 0_2^+
prompt	E_k [MeV]	1.404	2.96
	Γ_k [MeV]	1.4×10^{-4}	2.77
	$u_2(R_b = 8.5 \text{ fm})$		-0.047
	$u_2(R_b = 7.5 \text{ fm})$		-0.061
	$(s_{1/2})^2$ [%]	8.5	70.1
mixed	$(d_{5/2})^2$ [%]	88.4	5.7
	E_k [MeV]	1.401	2.44
	Γ_k [MeV]	1.1×10^{-3}	0.82
	$u_2(R_b = 8.5 \text{ fm})$		-0.109
	$u_2(R_b = 7.5 \text{ fm})$		-0.140
	$(s_{1/2})^2$ [%]	32.0	61.2
	$(d_{5/2})^2$ [%]	63.5	33.7



- In this work, **the 2p-radioactive emission** from the meta-stable 0^+ state of ^{16}Ne is investigated with **the time-dependent three-body model**.
 - By observing the time-dependent decaying state in the prompt case, a dominance of true emission is confirmed. This is a natural product of the energy scheme with the high $s_{1/2}$ resonance.
 - In the mixed case by assuming the intermediate $s_{1/2}$ resonance, the sequential emission appears as expected.

- The 0_2^+ resonance is solved by using the technique of **modified confining potential**, where the initial state can have a finite inclusion of the 0_2^+ resonance.
 - In the prompt case, the 0_2^+ resonance is predicted at $\sim 3\text{MeV}$ from the $2p$ threshold.
 - In addition, by including this 0_2^+ resonance, the true $2p$ emission becomes more dominant.

- ❑ Search for certain emission process, where **the predicted 0_2^+ resonance** can be apparent, is in progress now. One possible case was reported in **the pionic double charge-exchange reaction**, where its energy is $\sim 1\text{ MeV}$ higher than the result in this paper.

Thank you for your attention!